Documentation interfaces “interfaz_remix”

This document contains the description of the interfaces used in the file “main_interfaz.f90”.

Note: the variables that contain an asterisk are **optional** arguments.

`read_chemistry`

- **Description:** Interface to read all the chemical information (species, reactions, initial and boundary concentrations, etc).
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
`root`	String	Root of input and output files of problem to be solved
`path_pb`	String	Path to input and output files of problem to be solved
`path_DB`	String	Path to chemical databases

- **Pre-process:**
    - Sets arbitrary input and output file units
- **Process**:
    - Reads **chemical options** calling subroutine `read_chem_opts`
    - Reads **chemical data** depending on the format (CHEPROO-based, PHREEQC, PFLOTRAN). Currently, only CHEPROO-based option is available.

`get_num_aq_comps`

- **Description:** Gets the number of aqueous components in a reactive zone. The index of the reactive zone is optional. If no index specified, it assumes that all reactive zones have the same number of aqueous components.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
`ind_rz*`	Integer	Index of reactive zone in chemistry object.

- **Output:**

Variable	Type	Description
`num_aq_comps`	Integer	Number of aqueous components in reactive zone.

- **Process**:
    - Checks if optional argument is present.
        - If present:
            - returns an error message if index is out of bounds. If not, returns number of aqueous components in `ind_rz`-th reactive zone.
        - If not present: returns number of aqueous components in first reactive zone.

`interfaz_comps_arch`

- **Description:** Interface to solve a reactive mixing iteration with components, using **files** as **arguments**. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
`path`	String	Path to input and output files.
`num_aq_comps`	Integer	Nº aqueous components.
`file_in`	String	File containing concentrations of aqueous components after solving conservative transport iteration $(\tilde{\mathbf{u}}^k)$.
`Delta_t`	Real	Time step $\left(\Delta t^k\right)$
`file_out`	String	File where concentrations of aqueous components after solving reactive mixing iteration will be written $(\mathbf{u}^{k+1})$.

- **Pre-process:**
    - Allocate aqueous component concentrations after conservative transport $(\tilde{\mathbf{u}}^k)$
    - Allocate chemical reactions contribution to aqueous component concentrations $\left(\mathbf{u}_K^k\right)$
- **Process**:
    - **Read** $\tilde{\mathbf{u}}^k$ from `file_in`. **Note:** The rows must represent components and the columns targets.
    - **Solve** reactive mixing iteration for each target water:
        - **Loop over number of target waters in domain**
            - Compute chemical reactions contribution to aqueous component concentrations using **Euler explicit** and **lumping**:
            $$\mathbf{u}_{K,j}^k := \Delta t^k \mathbf{U}\mathbf{S}_{K,nc}^T \mathbf{r}_{K,j}^k$$
            - Update the total aqueous component concentration in target water: $\mathbf{u}_j^{k+1} = \tilde{\mathbf{u}}_j{}^k + \mathbf{u}_{K,j}^k$
- **Post-process:**
    - **Write** $\mathbf{u}^{k+1}$ in `file_out` (same dimensions as $\tilde{\mathbf{u}}^k$)
    - **Deallocate** arrays